EV Charging VR Research Audit

Oct 23, 2026 (rev. 4) · @Naman Khatak

A grounded audit of the EV-charging cybersecurity VR project (BeamNG.tech + Varjo XR-4 + Muse S Athena): what the codebase and data actually show versus what the mid-sem report and literature review claim, and a research plan built on that evidence.

What's new in this revision. Revision 3 (Oct 3) left the deep-learning pipeline, the confirmatory statistics, and the AI-warning mechanism as open design items. This revision closes all three with real, run numbers: classical ML baselines and a gated MLP check (D037), a critical labeling-bug fix that changed the base risky-rate from 19.1% to 34.7% (D045), a consolidated CI-backed ablation table superseding every earlier number (D050), a confirmatory mixed-effects model with FDR-corrected gaze-feature effect sizes (D052), a deployed risk model with calibrated probability thresholds (D055/D057), real-time (streaming) gaze-feature computation (D056), and a git history now published for the project (D053, D058). It also surfaces the single most important new finding: **the deployed model does not match the architecture this audit and the mid-sem report describe** - see "Architecture reconciliation" below. Every claim below is traceable to a dated, numbered Decision.md entry (D033-D058); this document restates and contextualizes them, it does not replace that log as the source of truth.

The project (MTP-1, IIT Mandi, Arpita Kumari + Naman Khatak, supervisor Prof. Varun Dutt, mentor Ranik Goyal) studies how people make decisions at compromised EV charging kiosks inside a VR driving simulation, and whether behaviour, gaze and EEG can flag the risky ones. The participant drives an EV in BeamNG.tech with a 400-point budget, reaches a station of three kiosks, and can charge, skip, or ask an in-sim AI assistant for help. Some kiosks are malicious: a fake discount coupon, incomplete charge delivery, or overcharging. Eye movements are captured with a Varjo XR-4 (built-in eye tracking) and EEG with a 4-channel Muse S Athena.

Objective status

Treating every objective as uncompleted unless the code, the data on disk, or a real run shows otherwise.

Objective	Current status	Required work	Evidence
1. FPS optimization	Unchanged since rev 3: application-layer config fixes applied (shadow/reflection scalars); engine itself is a closed compiled binary, cannot be profiled by reading code.	Run a Windows-side profiler pass on the lab workstation; measure before/after FPS.	rev 3 Objective 1 (unchanged - no Windows access this revision); Decision.md pre-D033
2. AI pipeline + warnings	Classical ML baselines + gated MLP check run for real (D037); critical labeling bug found and fixed, changing the base risky-rate from 19.1% to 34.7% (D045); consolidated CI-backed ablation table, participant-level bootstrap (D050); confirmatory mixed-effects model with FDR-corrected gaze-feature odds ratios (D052); risk-decision model + AI-warning scaffold built, calibrated, and unit-tested (D055/D057); real-time streaming gaze-feature computation built (D056). **Does not match the originally proposed architecture** - see "Architecture reconciliation".	Video-feature extraction (never implemented); LSTM/GRU and Transformer phases (gated off, not attempted, since the MLP never beat the classical baseline); a 5-SLM-vs-ML classification comparison (script written, deferred to the lab PC - D058); live-wiring the warning system to an actual real-time gaze/video stream.	backend/pipeline/train_baselines.py, train_mlp.py, final_ablation_table.py, mixed_effects_model.py, calibrate_risk_model.py; backend/risk_model.py, warning_system.py, gaze_online.py; Decision.md D037, D045, D047-D050, D052, D055-D058

Objective	Current status	Required work	Evidence
3. New data collection	Still not started (correct per report). Infrastructure prepped without touching the lab workstation: RAG-store auto-clear fixed and unit-tested (D053); head-pose Lua logging hook written (D053, marked UNTESTED - never loaded into a real BeamNG instance); Python-side consumer for that hook written and unit-tested on pure data-handling logic (D054); per-session QC script (D053); collection protocol document (D053).	Load the Lua hook into a real BeamNG instance and confirm the buffer round-trips (lab-only); run the calibration-nod protocol to measure the real Lua-to-Python clock offset (lab-only); locate the external gaze-capture application and the flowgraph scenario file (lab-only, rev 3's standing blocker, unchanged).	BeamNG.tech.../lua/ge/extensions/evHeadPoseLogger.lua, backend/receive_ui.py (_headpose_worker), backend/pipeline/session_qc.py, NEW_DATA_COLLECTION_PROTOCOL.md; Decision.md D053, D054
4. Literature review & gap analysis	Unchanged since rev 3 - substantially complete, unusually rigorous for this stage.	Verify the four citations still flagged [to verify] in the report's own reference list (not done this revision).	rev 3 Objective 4 (unchanged); mid-sem report References section

The most consequential correction in this revision sits inside Objective 2, and it is new, not carried over from rev 3: the mid-sem report's own proposed pipeline (Figures 1-2, Section 10) fuses video + eye-tracking + EEG through a phased deep-learning model. What has actually been built and run uses behaviour + gaze only, stopped at the first (MLP) phase per the report's own gating rule. This is explained in full below rather than left as an implicit gap.

Architecture reconciliation (new this revision)

This section did not exist in rev 3. It compares the mid-sem report's own Figures 1-2 (the proposed pipeline) against what the codebase and Decision.md show has actually been run, line by line, so the gap is explicit rather than discoverable only by reading both documents separately.

Proposed (report Section 10)	Actually built/run	Status
Feature-level fusion of video + eye + EEG into one vector (Fig. 2)	Deployed model uses behaviour (minus the RAG-contamination flag) + 6 AOI-free gaze features only. No video features exist anywhere in the codebase. EEG was evaluated in the ablation but excluded from the deployed model.	Deviation - traced to real, dated findings: EEG qualifies for only 67/140 participants (D019) and adds no detectable signal over gaze+behaviour at this sample size (D050); video feature extraction (report Section 10.2) was never implemented.
Phase 1: MLP baseline	Run for real against the classical-baseline gate (D037): behaviour+gaze MLP scored balanced_acc=0.497/AUROC=0.522 (n=1057), below the 0.632/0.712 baseline it needed to beat on both metrics.	Done, and correctly failed its own gate.
Phase 2: LSTM/GRU (temporal)	Not attempted.	Correctly gated off - the report's own rule is "proceed only if the prior phase beats the one before it"; the MLP did not, so this was never supposed to run yet.
Phase 3: Transformer (multimodal attention)	Not attempted.	Correctly gated off, same reason as Phase 2.

Proposed (report Section 10)	Actually built/run	Status
Step 12: AI warning, SLM phrases only, never decides	Built exactly as specified: RiskModel (RandomForest, calibrated) makes the decision; WarningSystem applies a threshold; an SLM (local Mistral-7B via Ollama) only phrases the resulting message, structurally restricted to (risk_level, kiosk_label, reason) - never raw features, checked via inspect.signature in the test suite.	Matches exactly - the one part of the proposed pipeline implemented as designed.

Why this matters going forward, stated plainly: the classical-ML half of the proposed pipeline (baselines through the MLP gate) has been executed faithfully and honestly - including reporting that the MLP lost, rather than skipping it on the assumption it would. The deep multimodal half (video fusion, LSTM/GRU, Transformer) has not been attempted at all, not because it was tried and failed, but because video features were never built and the MLP never cleared the gate that would unlock the next phase. Framing the current RandomForest as "the pipeline" without this context would overstate what exists; framing the gap as neglect would understate that every stopping point was a measured, reported gate, not a silent shortcut.

Objective 1 - FPS optimization (unchanged since rev 3)

No new FPS work this revision - this remains blocked on the Windows/lab workstation, which this Mac has no access to. Restating rev 3's findings for continuity: BeamNG.tech v0.38.3.0 is a closed commercial product here, not an open codebase - Bin64/ holds compiled .exe/.dll files and gameengine.zip is assets-only. What is inspectable and genuinely "ours": the Lua gameplay layer, the custom EVChargingGame BeamNG UI app, and the Python backend that drives it via BeamNGpy. Config-level fixes (Shadows::disable, Shadows::textureScalar, Reflect::refractTexScale) were applied but never measured from this Mac - that measurement still requires the Windows workstation. The Varjo XR-4 panel refreshes at 90 Hz, so the report's 120 FPS target should be revised to a stable 90.

Objective 2 - results (all run for real, participant-level splits throughout)

Labeling bug correction (D045). The risky-decision label originally matched only the string "Cyber Attack", silently mislabeling every session logged under the equivalent "System Anomaly" vocabulary (56/138 sessions, 15.7% of the dataset) as safe. Fixed to match either label for the identical underlying attack mechanic. Corrected overall risky-decision rate: **34.7% (390/1123)**, not 19.1% as every result before this fix reported.

Final, CI-backed ablation table (D050) - supersedes every ablation number in D037-D045. Selected rows (full 16-cell table in Decision.md):

Feature set	Protocol	n (participants)	Balanced acc. [95% CI]	AUROC [95% CI]
Behaviour-only	time-ordered	316 (32)	0.599 [0.533, 0.659]	0.690 [0.615, 0.754]
Gaze-only	time-ordered	308 (31)	0.625 [0.567, 0.685]	0.676 [0.605, 0.740]
Behaviour+gaze (no rag_contaminated) - deployed config	time-ordered	308 (31)	0.620 [0.556, 0.682]	0.695 [0.623, 0.759]
Behaviour+gaze+EEG (no rag_contaminated)	time (EEG window)	87 (14)	0.651 [0.573, 0.707]	0.695 [0.628, 0.749]

The behaviour+gaze+EEG AUROC CI [0.628, 0.749] overlaps almost completely with behaviour+gaze alone [0.623, 0.759] - both are genuinely above chance, but EEG's incremental contribution is not resolvable at this sample size. This is why the deployed production model (D055) uses behaviour+gaze only, not all three modalities.

Confirmatory mixed-effects model (D052). The literal specification (*risky_decision ~ attack_type + (1|participant) + (1|station)*, using the kiosk the participant actually chose) turned out to be unfittable - complete separation by construction, since a risky decision is defined as charging at a kiosk already known to be compromised. Reported as a genuine finding rather than forced. A non-circular version (station-level attack presence, independent of the participant's choice) fit successfully via statsmodels' BinomialBayesMixedGLM. **Do not headline its OR=67.7 [58.8, 77.9] effect** - it is close to a tautology, since a station with no compromised kiosk cannot produce a risky outcome, and is reported here only as a sanity check. The citable findings are the FDR-corrected AOI-free gaze-feature effects, each added individually to the model:

Gaze feature (z-scored)	OR	95% CI	FDR q	Survives q<0.05?
gaze_n_clusters	1.66	[1.44, 1.92]	<0.0001	Yes

Gaze feature (z-scored)	OR	95% CI	FDR q	Survives q<0.05?
gaze_entropy_bits	1.32	[1.14, 1.52]	0.0002	Yes
gaze_transitions_per_fixation	0.87	[0.76, 1.00]	0.0533	No (CI includes 1.0)

More gaze clusters and higher gaze entropy - both D025/D030-validated, AOI-free features - are each independently associated with higher odds of a risky decision, surviving FDR correction across the three gaze features tested.

Risk-model calibration (D057). The deployed RandomForest's raw probability output was checked against held-out, participant-level out-of-fold probabilities and found non-monotonically miscalibrated in the upper range. Two calibrators were compared via nested cross-validation: isotonic regression achieved a better Brier score (0.2067 vs 0.2153) but compressed 99% of its output below ~0.48, unusable for a "high" threshold. Platt/sigmoid scaling was deployed instead, explicitly despite the worse Brier score, because its output stays smoothly spread. New thresholds (low=0.25, high=0.50) were set only after inspecting the real calibrated distribution. Measured operating point on held-out data: 22.9% low, 10.5% high, 77.1% would trigger a warning - a direct, honest consequence of the model's AUROC (~0.6-0.7), not a tuning bug.

Real-time gaze-feature computation (D056). Closes the "no live feature computation exists" gap this audit would otherwise still be flagging: GazeOnlineProcessor recomputes the same AOI-free cluster features the offline pipeline uses, validated for exact agreement against the offline computation on real sessions, with latency comfortably inside a 5Hz polling budget on this Mac (mean 0.645ms, max 24.7ms tail). Still not wired to any live gaze stream - D027's finding stands: no connection from the external gaze-capture application into this project's backend has ever been found or built. This closes the feature-computation half of the live-warning gap, not the data-source half.

5-SLM-vs-ML classification comparison (D058, deferred). A script (compare_slm_classifiers.py) was written to compare five local SLMs (Mistral, Llama 3.1, Phi-3, Gemma2, Qwen2.5), zero-shot, against the deployed RandomForest on the same held-out fold and feature set. Smoke-tested on 5 rows only - no real evaluation exists yet. Deferred to the lab PC, which has the GPU and the full video dataset this comparison could meaningfully extend to.

Objective 3 - new data collection (infrastructure prepped, collection not started)

All of this was built without touching the lab/Windows workstation, and nothing here constitutes actual new data collection - that remains Not Started, correctly, pending the lab-only items listed below.

- RAG-store cross-session contamination (flagged in rev 3 as affecting all 38 sessions that used the assistant) is now fixed going forward: AIAssistant.reset_session() calls RAGStore.clear() (D053). The old contaminated index was archived, not deleted. Unit-tested.
- Head-pose logging: a BeamNG Lua extension (evHeadPoseLogger.lua) buffers core_camera position/orientation per tick plus a calibration-nod marker (D053) - explicitly marked UNTESTED in its own header, never loaded into a real BeamNG instance. A Python-side consumer (receive_ui.py's _headpose_worker, D054) drains it and reconstructs timestamps from the existing wall-clock convention - 8 unit tests pass on the pure data-handling logic, which is everything testable without the real workstation.
- Per-session QC script (session_qc.py, D053) and a one-page collection protocol document (NEW_DATA_COLLECTION_PROTOCOL.md, D053), including the recommendation to record incident_mode explicitly per session, since D045 found it silently changes the attack-label vocabulary for the identical underlying mechanic.

Still blocked on the lab workstation, not more code: loading the Lua extension into a real running BeamNG instance and confirming the buffer round-trips through queue_lua_command; running the calibration-nod protocol to measure the real Lua-to-Python clock offset; locating the external gaze-capture application and the flowgraph scenario file (rev 3's standing blocker, unchanged this revision - not resolvable from this codebase alone).

Objective 4 - literature review (unchanged since rev 3)

No new domain search performed this revision. The literature review document remains unusually rigorous for this stage, covering seven domains with a structured evidence matrix. Four citations remain flagged [to verify] in the report's own reference list - verifying them is still an open, Mac-doable task, not completed this revision.

Project infrastructure (new this revision)

- The project now has a published git history: github.com/namankhatakdotcpp/ev-cybersecurity-mtp (public). .gitignore excludes the vendored BeamNG engine, the Python venv, raw_data/, decoded EEG, and secrets/.env - only source code, Decision.md, and project docs are tracked.
- A README.md summarizes the project for anyone opening the repo cold: problem statement, the architecture-reconciliation table above, the citable results table, repo layout, and how to run the pipeline and tests.
- Workflow going forward splits across two machines (D058): this Mac for literature/design/doc work that needs no GPU or video data; the lab PC (full video dataset, RTX 4090) for the SLM comparison, any future deep-learning phases, and all of Objective 3's remaining physical-hardware items. Decision.md is the one thing both machines share via git, and stays the authoritative source of truth across both.

Research gap (unchanged since rev 3)

Area	Existing research	Weakness of prior work	Opportunity here
EV cybersecurity	Extensive, technical (Nasr et al. pentested 16 real systems)	No human subjects anywhere in the technical literature	First human-in-the-loop EV-charging deception study
Eye tracking + security	Moderate (phishing)	Attention != accuracy; static AOIs only	Dynamic 3D VR AOIs, decision-linked gaze - gaze data exists for all 140 sessions
EEG + eye-tracking fusion	Small, fragmented (18 papers per Thomopoulos et al.)	Lab-static settings, small n	Answers the field's own call to extend into a realistic multi-sensory environment
EV + human factors	Essentially absent	-	The core, well-evidenced gap

Full seven-domain table unchanged from rev 3 - see that revision or the literature review document for the remaining rows.

Professor presentation summary

A one-page version of this audit's load-bearing findings, for the next meeting with Prof. Dutt and Ranik Goyal.

- **1. (New) The deployed pipeline does not match the proposed architecture, and that gap is now documented explicitly, not implicit.** The report's own Figures 1-2 call for video+eye+EEG fusion through MLP -> LSTM/GRU -> Transformer phases. What exists: a behaviour+gaze RandomForest, with the MLP phase run for real and correctly failing its own gate (D037) - so LSTM/GRU and Transformer were never supposed to run yet, not neglected. Video features were never built at all.
- **2. (New) The risky-decision base rate was wrong in every result before this round - 19.1% vs. the corrected 34.7% (D045).** A labeling bug matched only one of two equivalent attack vocabularies. Every downstream number has since been recomputed; the consolidated table (D050) is the one source to cite going forward.
- **3. (New) A confirmatory mixed-effects model now exists, with a caveat worth stating out loud in the room before anyone else does.** Its headline odds ratio (67.7) is close to a tautology; the citable findings are two FDR-surviving gaze-feature effects (cluster count OR 1.66, entropy OR 1.32) - real, modest, and defensible.
- **4. (New) The AI-warning scaffold is built, calibrated, and unit-tested - but it is a scaffold, not a live system, for a specific, checked reason.** No real-time gaze-feature computation existed until this round (now fixed, D056); no live gaze stream has ever been found connecting the external capture application to this project's backend (D027, still unresolved). Any performance number cited for this model must come from D050's held-out CIs, not from scoring a session the production model was trained on.
- 5. FPS optimization is unchanged and still blocked on a Windows measurement that hasn't happened - same item rev 3 flagged as the one thing only lab access can unblock.
- 6. New data collection remains correctly Not Started; the remaining blockers (locating the gaze-capture app and flowgraph file, loading the head-pose hook into a real BeamNG instance) are lab-only, unchanged since rev 3.
- 7. The project now has a published, git-tracked history - useful for exactly this kind of progress review, and for the two-machine (Mac + lab PC) workflow going forward.